

Software for Mechanical Characterization and Design of Textile-Reinforced Concrete in Flexure

Devansh Patel, Chidchanok Pleesudjai, Barzin Mobasher

RILEM 2026 Spring Convention
13-17 April 2026, Ghent, Belgium



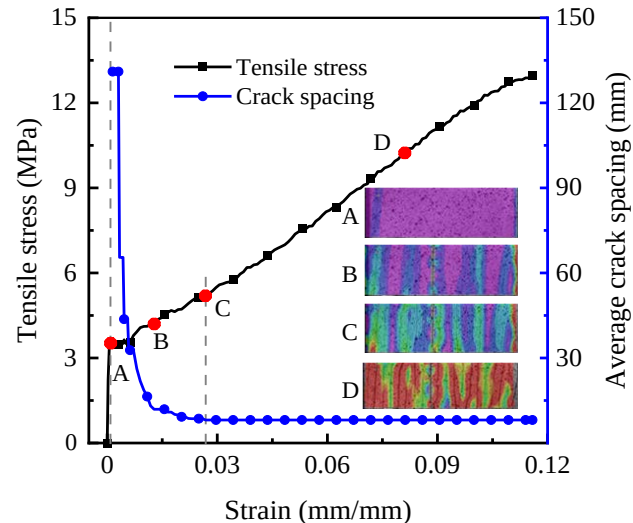
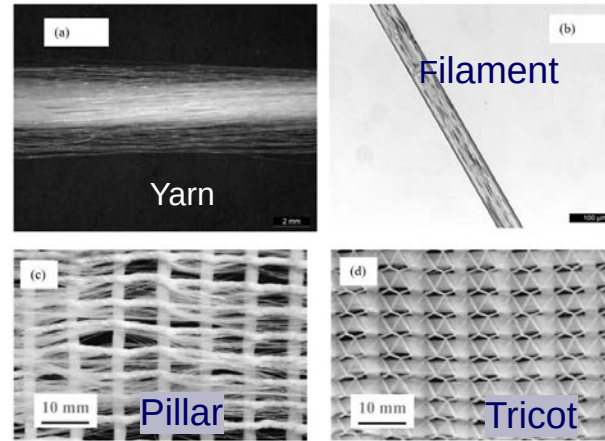
Distributed Cracking Mechanisms in TRC

- Warp knitting pattern (pillar vs. tricot) had no significant impact on tensile properties. Pillar knits recommended due to simpler, less labor-intensive stitching

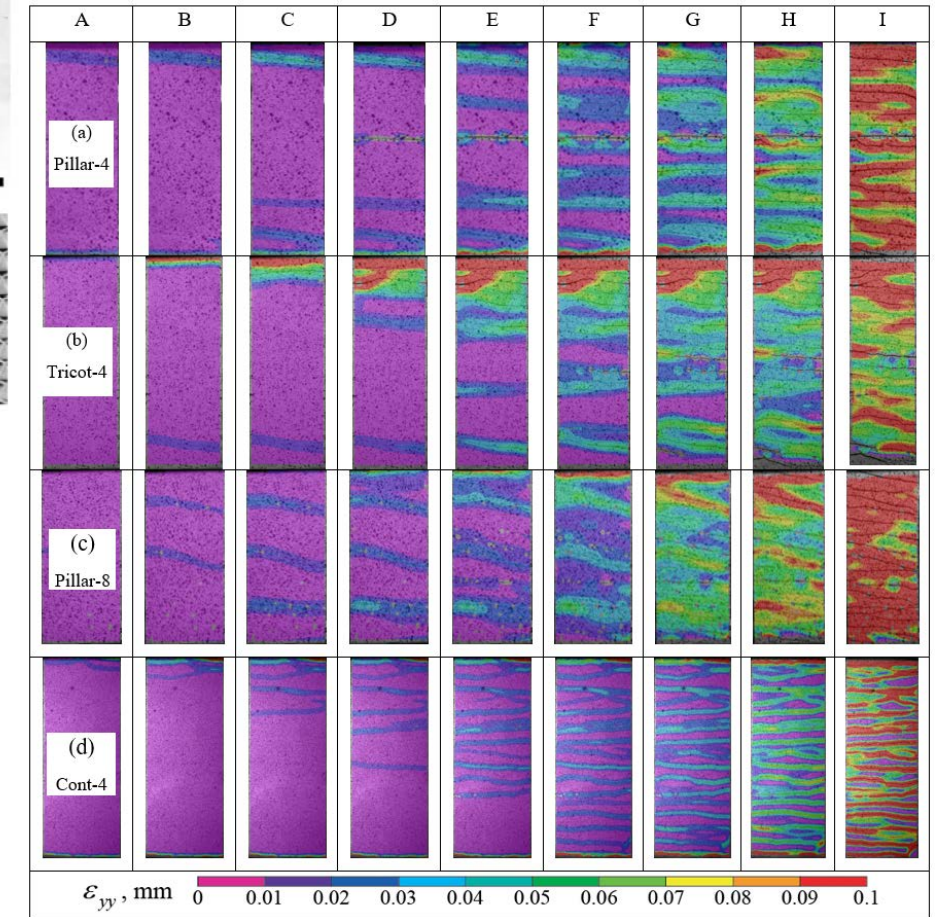
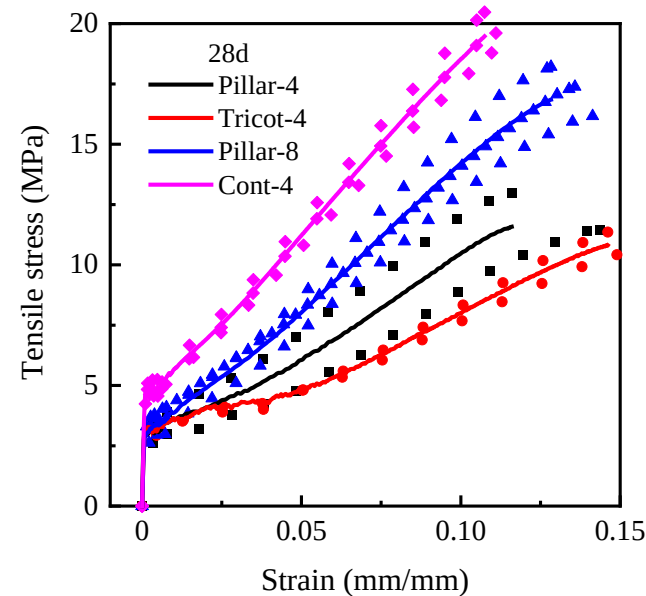
- Doubling textile volume fraction (4% to 8%) increased tensile strength by 44%, post-crack modulus by 78%, and toughness by 47%

- DIC revealed random crack formation rather than sequential spacing reduction along the gage length

- Crack width limit-state of 0.5 mm is satisfied at 75% of ultimate tensile strength (8–12 MPa depending on textile content)



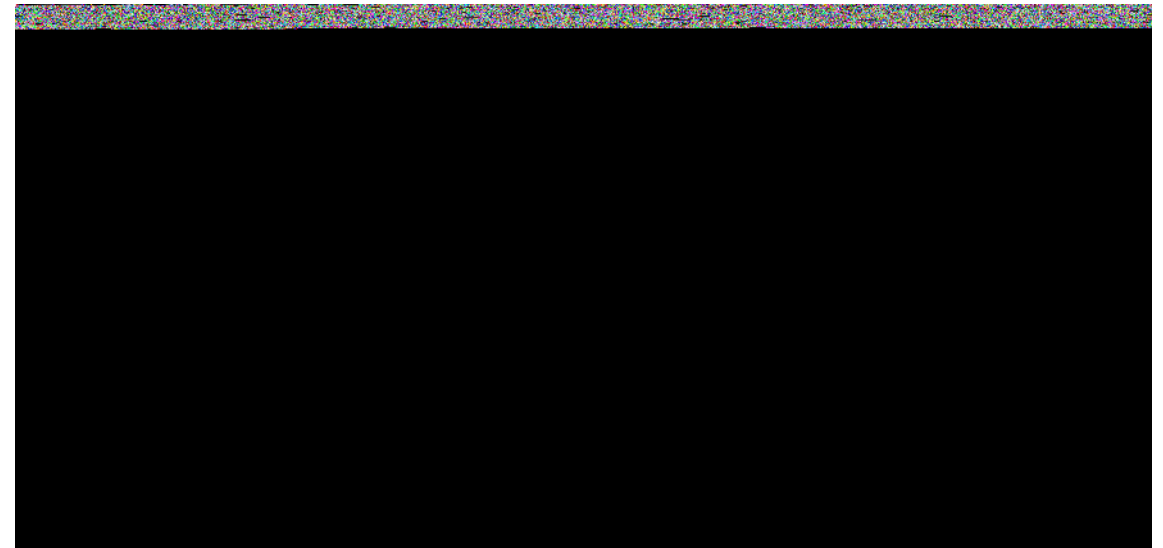
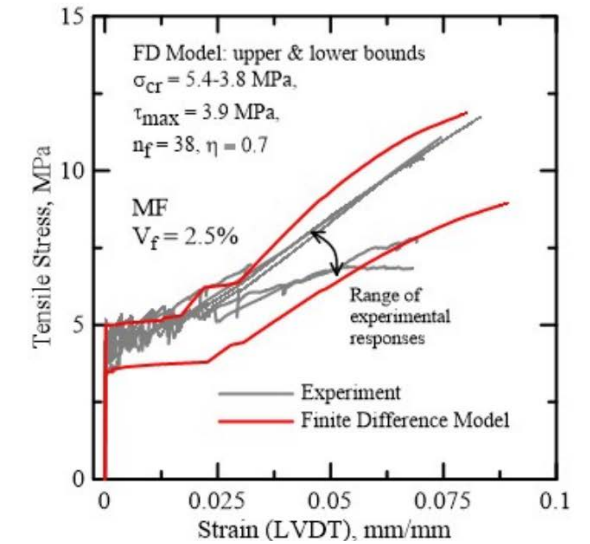
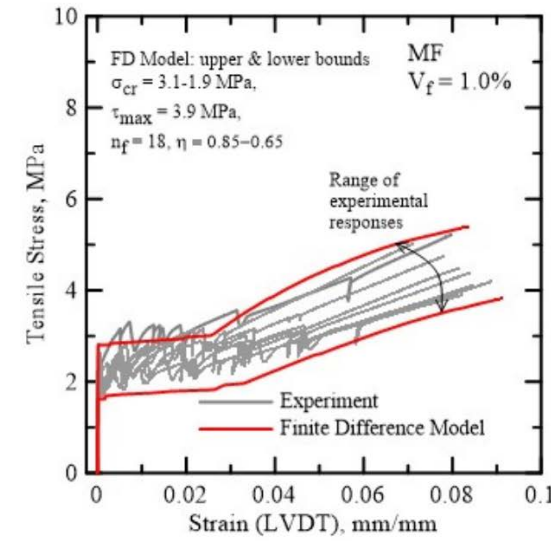
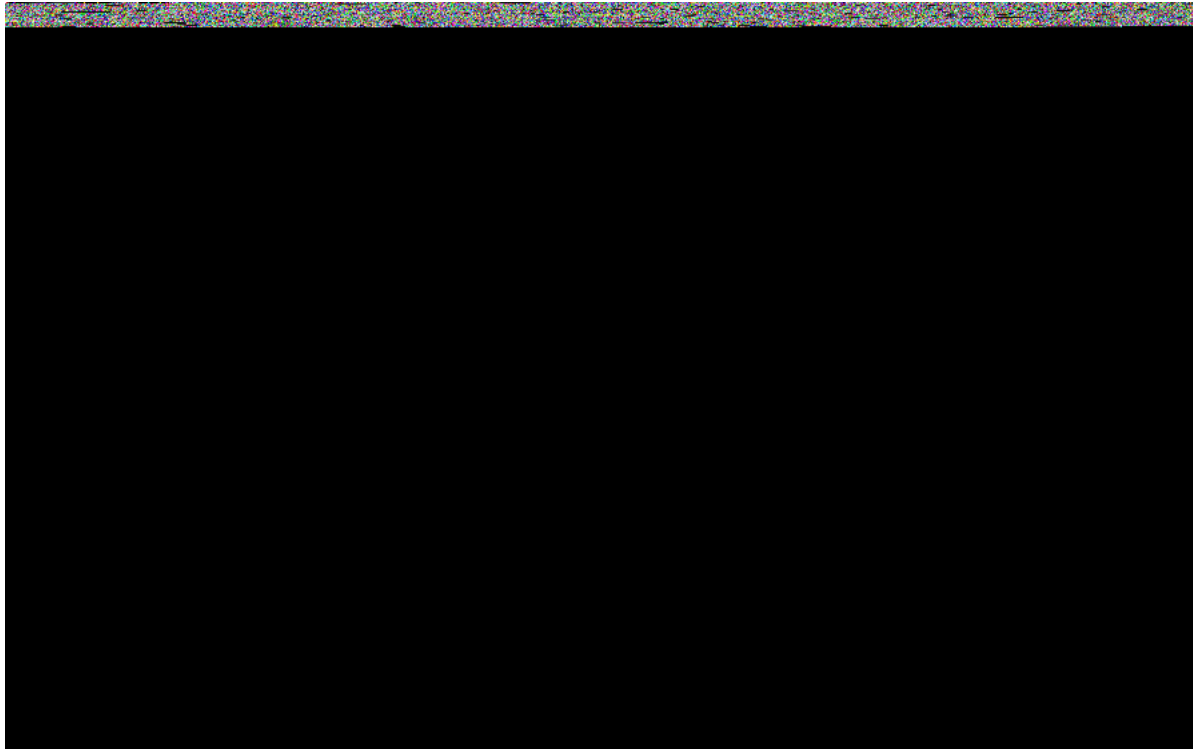
Crack spacing of pultruded pillar TRC specimen (Pillar-4): Stress-Strain vs. Crack location



- DIC tracks each crack via time stamp, strain, and image intensity.
- New cracks form randomly (not sequentially).

Distributed Cracking Mechanisms in TRC

3



Flexural Modeling with Parametric Approach

4

1) Constructing force equilibrium to calculate NA

$$\sum F = \varepsilon_{cr} E b \int_{-kh}^{(1-k)h} \sigma(\beta') dy = 0$$

$$0 = \frac{\varepsilon_{cr} E (1-k) b h}{\beta} \left[\int_{k(\beta)}^0 \sigma_c(\beta') d\beta' + \int_0^{\beta} \sigma_t(\beta') d\beta' \right]$$

It is a function of β and material parameters

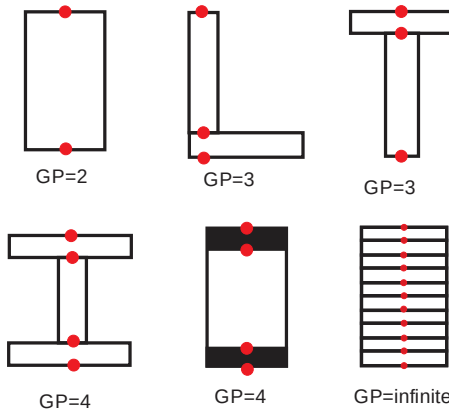
2) Substitute $k(\beta)$ in moment equation

$$M(\beta) = \int_0^h dM(y) = b \int_{(1-k)h}^{-kh} \sigma(\beta') y dy$$

$$= \frac{\varepsilon_{cr} E b h^2 (1-k)^2}{\beta^2} \left\{ \int_0^{\beta} (A_1 \beta' + B_1 \beta'^2) d\beta' + \int_0^{\lambda} (A_2 \beta' + B_2 \beta'^2) d\beta' \right\}$$

3) Determine moment at each stage and generate the envelope moment using the boundary condition for each stage

The stress distribution is based on linearized strain at different Stages.



● Geometrical control point, (GP)

“Closed-Form Equations”

$$\varepsilon_{1,top} = \lambda \varepsilon_{cr}$$

$$\varepsilon_{2,top} = \frac{k-1+\alpha_c}{(1-k)} \beta \varepsilon_{cr}$$

$$\varepsilon_{3,bot} = \frac{\alpha_t - k}{(1-k)} \beta \varepsilon_{cr}$$

$$\varepsilon_{4,bot} = \beta \varepsilon_{cr}$$

Intersection point connected with logarithmic function of rectangular section

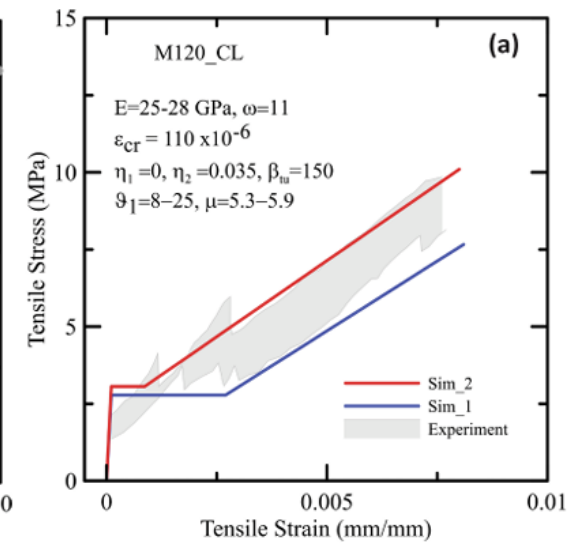
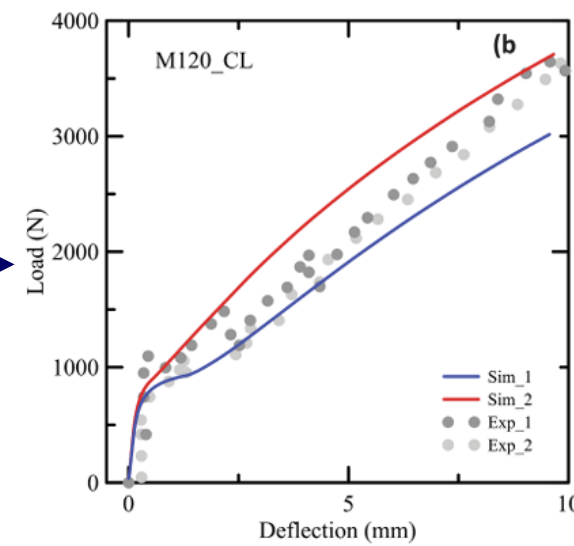
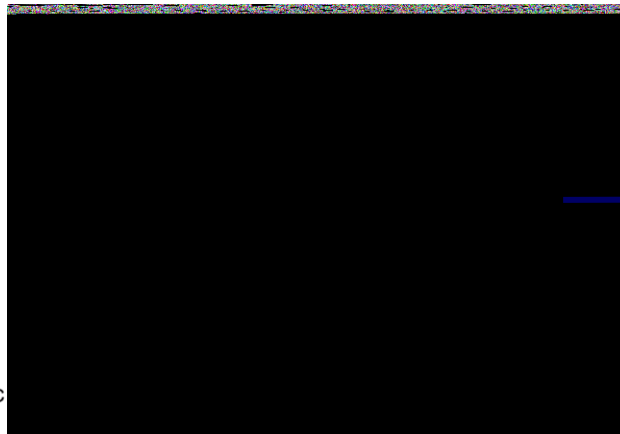
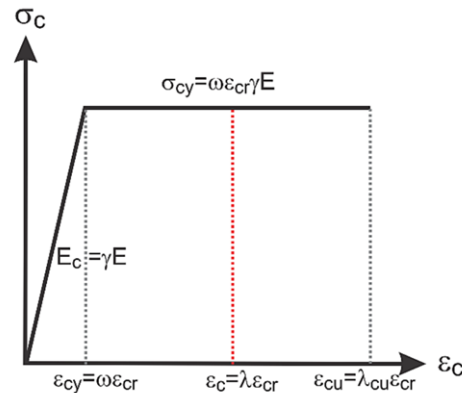
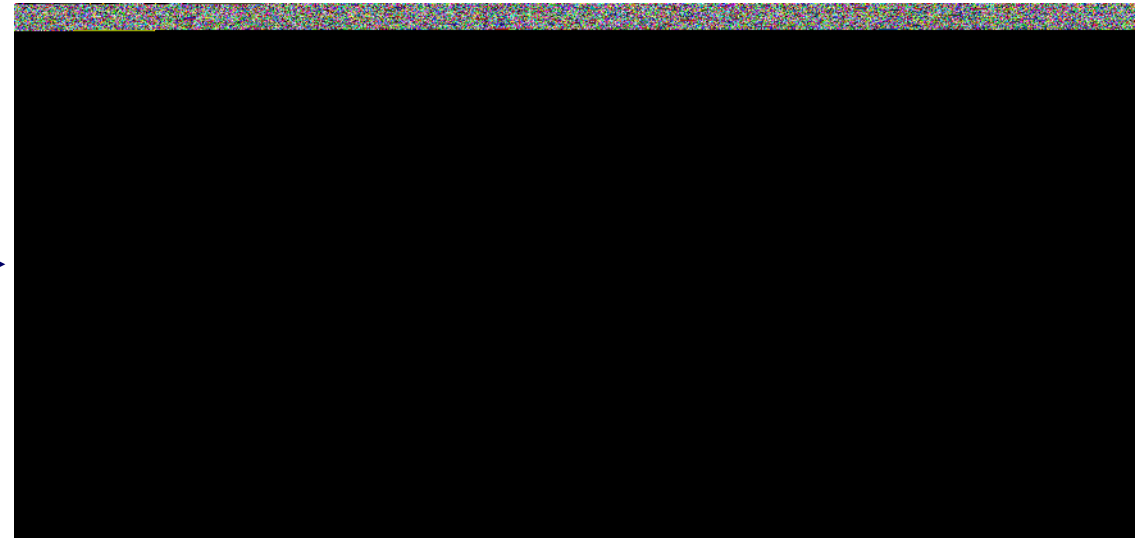
$$\beta' = \{\beta_{1,top}, \beta_{2,top}, \beta_{3,bot}, \beta_{4,bot}\} = \{\lambda, \frac{k-1+\alpha_c}{(1-k)}, \frac{\alpha_t - k}{(1-k)}, 1\} \beta$$

1. Pleesudjai, C., & Mobasher, B. (2023). Analytical moment-curvature solutions for generalized textile-reinforced concrete sections. *Engineering Structures*, 276(May 2022), 115317. <https://doi.org/10.1016/j.engstruct.2022.115317>

2. Mobasher, B., Pleesudjai, C., & Yao, Y. (2021, June). Hierarchical Modeling of Textile Reinforced Concrete Structural Systems. *The 13th International Symposium on Ferrocement and Thin Fiber Reinforced Inorganic Matrices*, Lyon, France

Application of Model Types

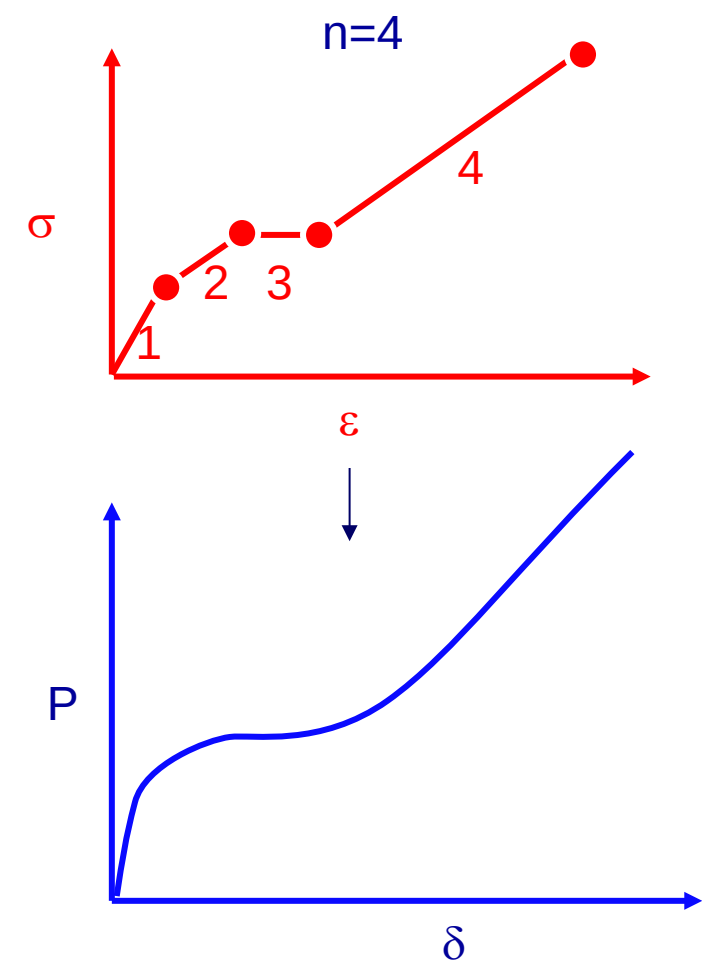
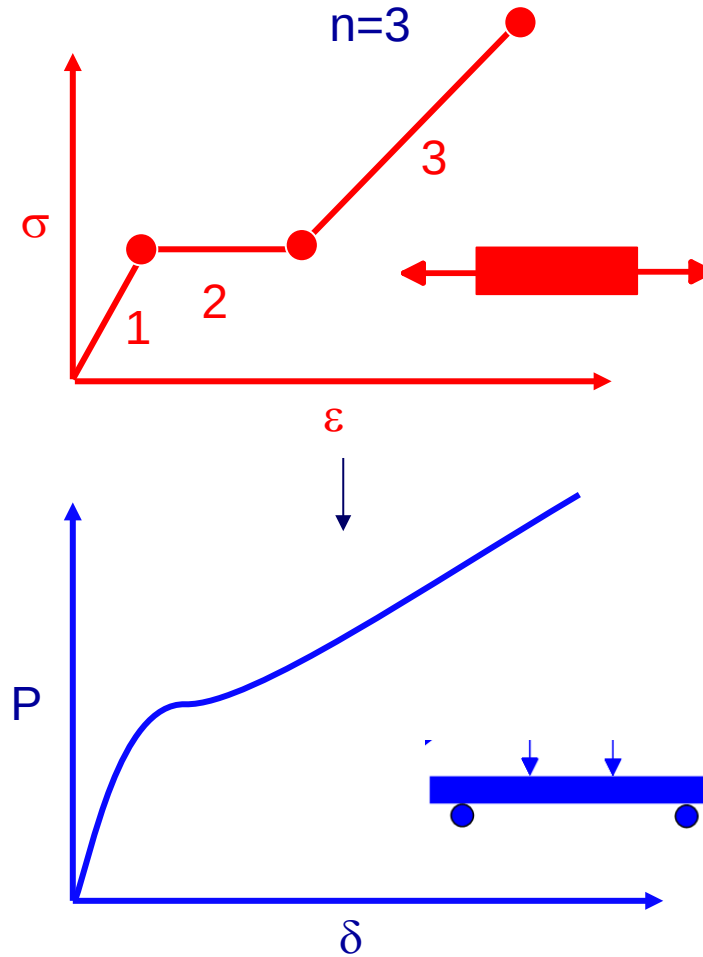
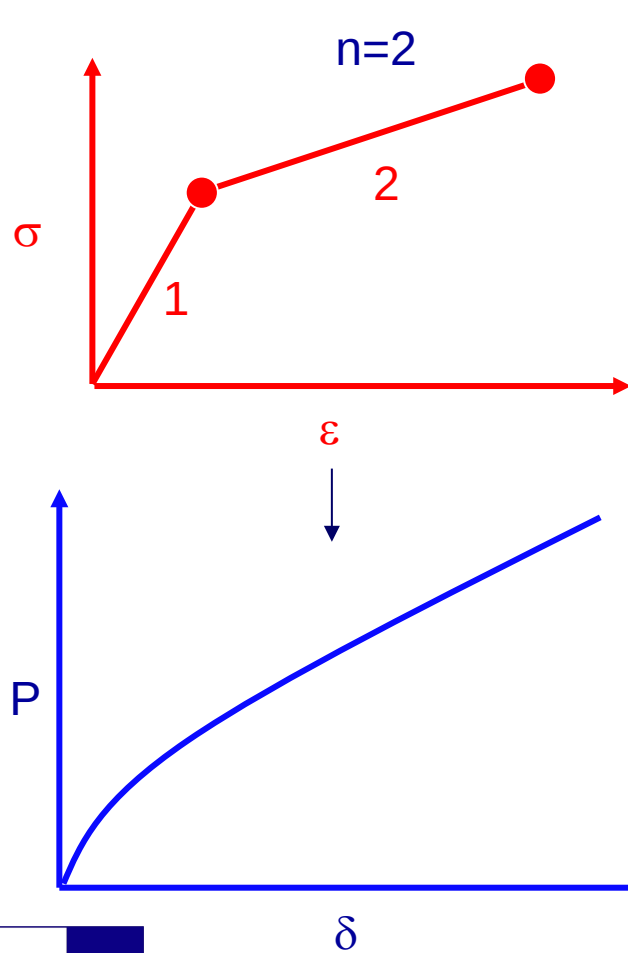
5



Constitutive Tension Model

6

Higher order parametric tension model



GUI Software Tool

7

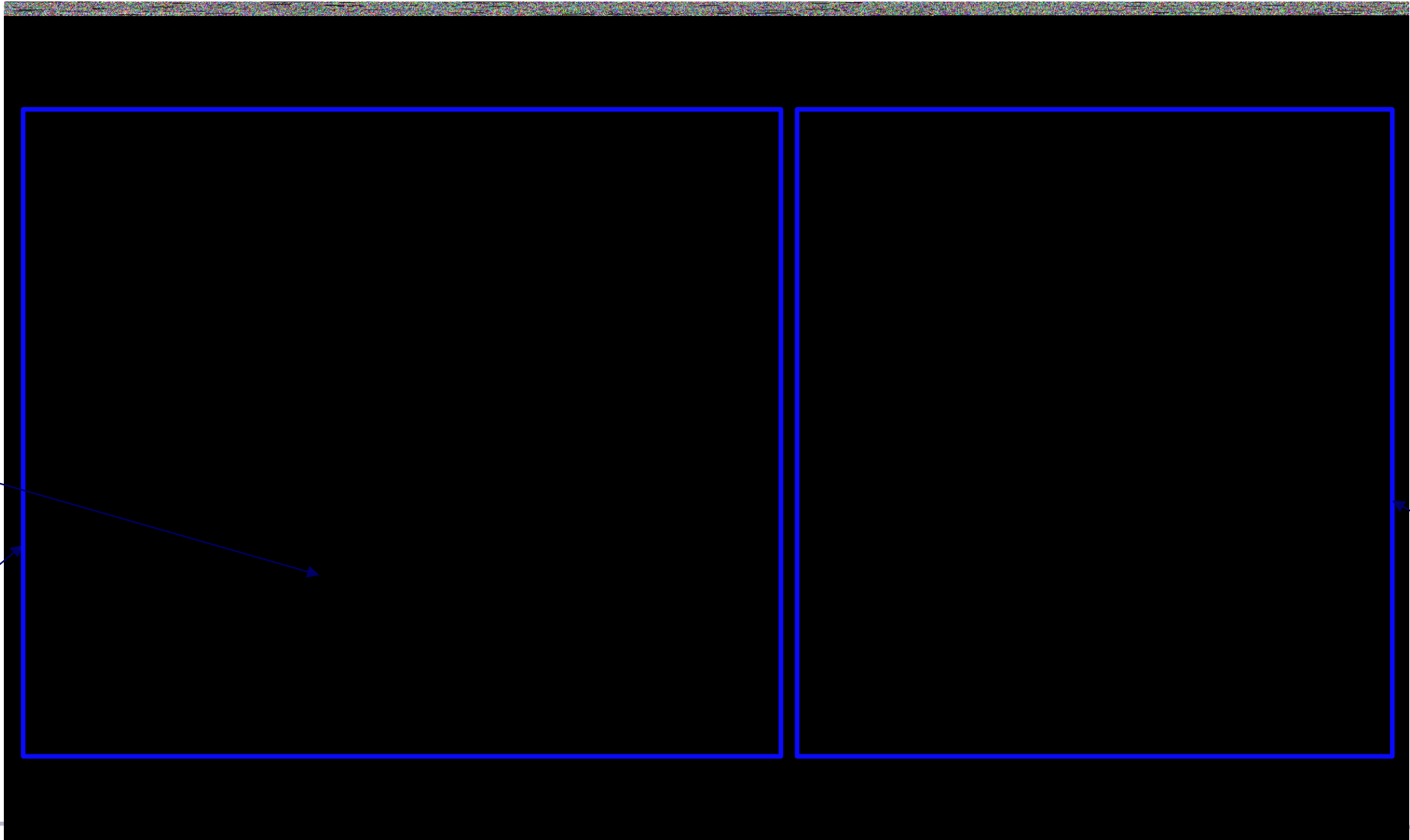
MATLAB based stand-alone software tool for concrete composites



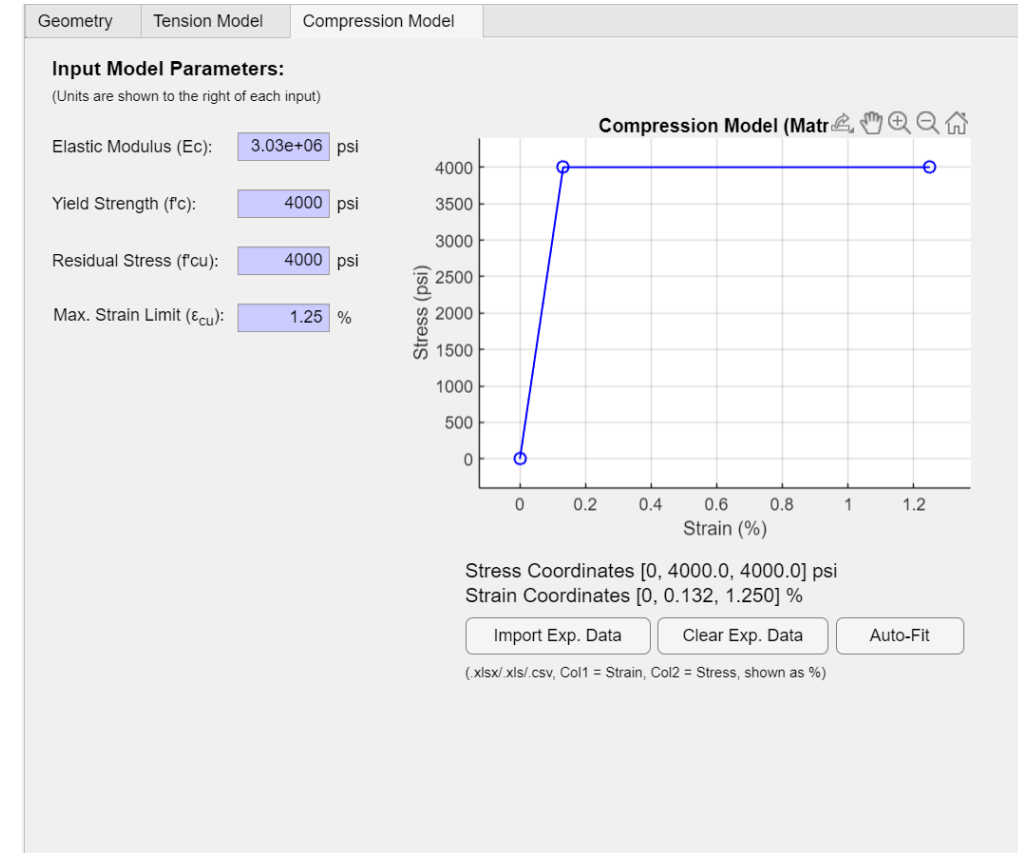
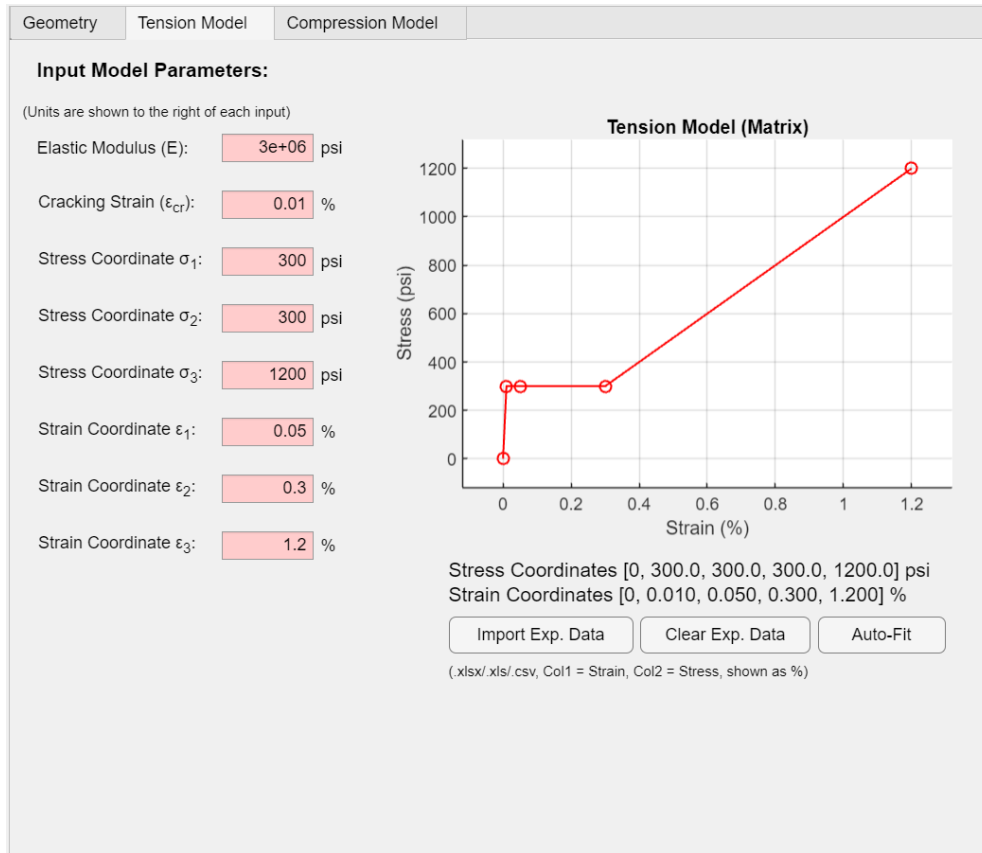
To-scale
graphics

Input
window

Output
window



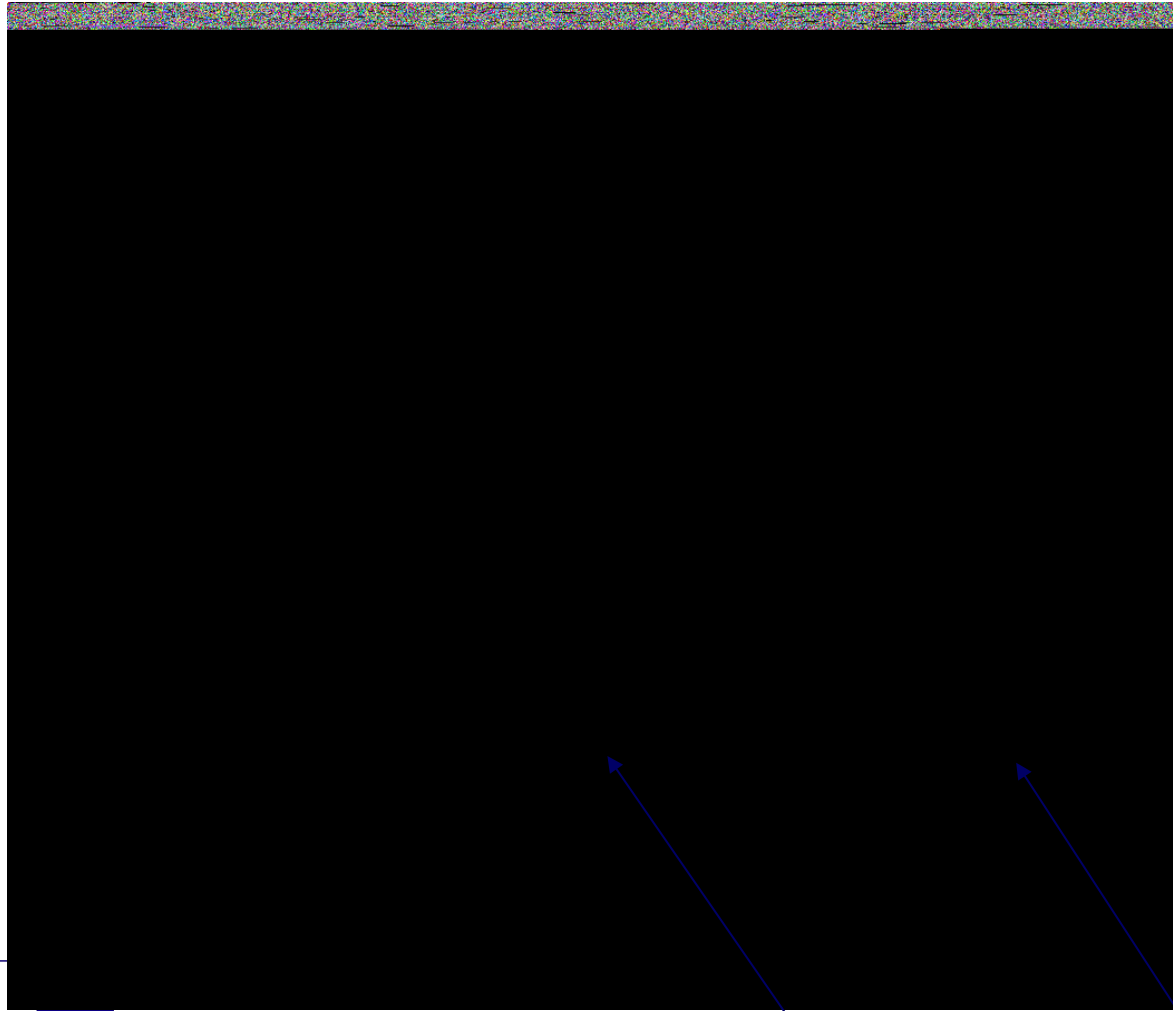
Material model window



GUI Software Tool for Forward Analysis

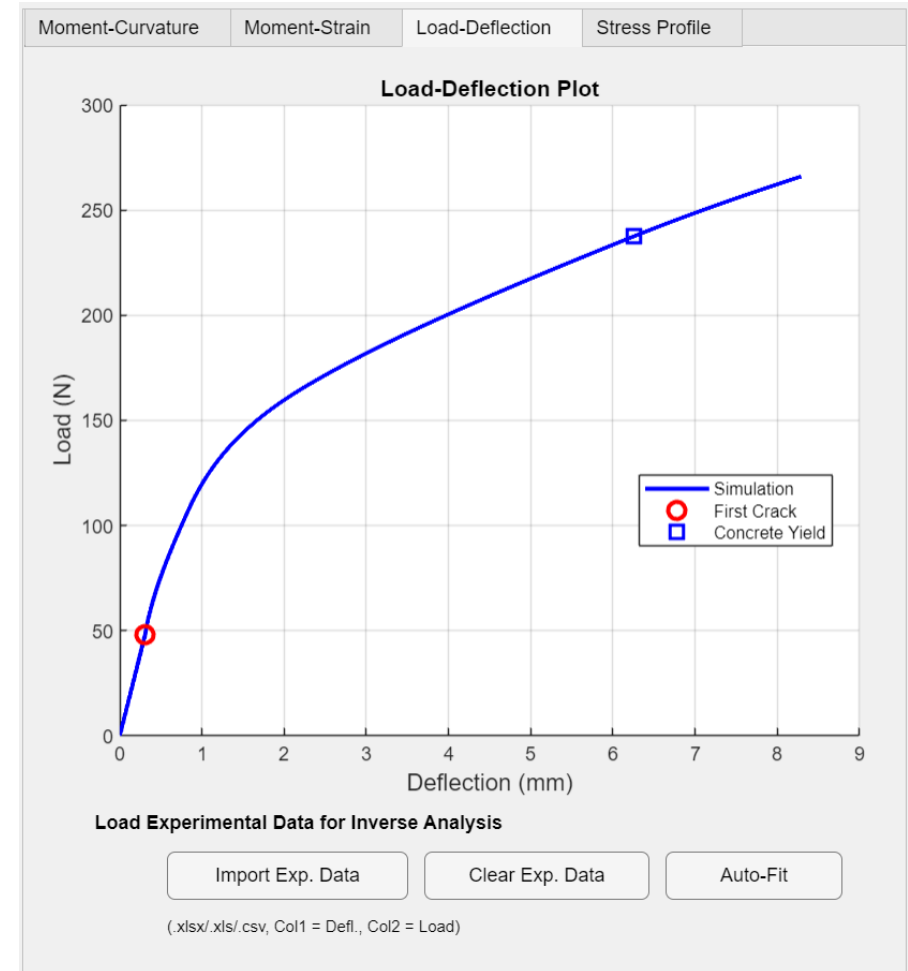
10

Forward-Analysis - Tension Model fitting



Direct upload

Load-deflection response

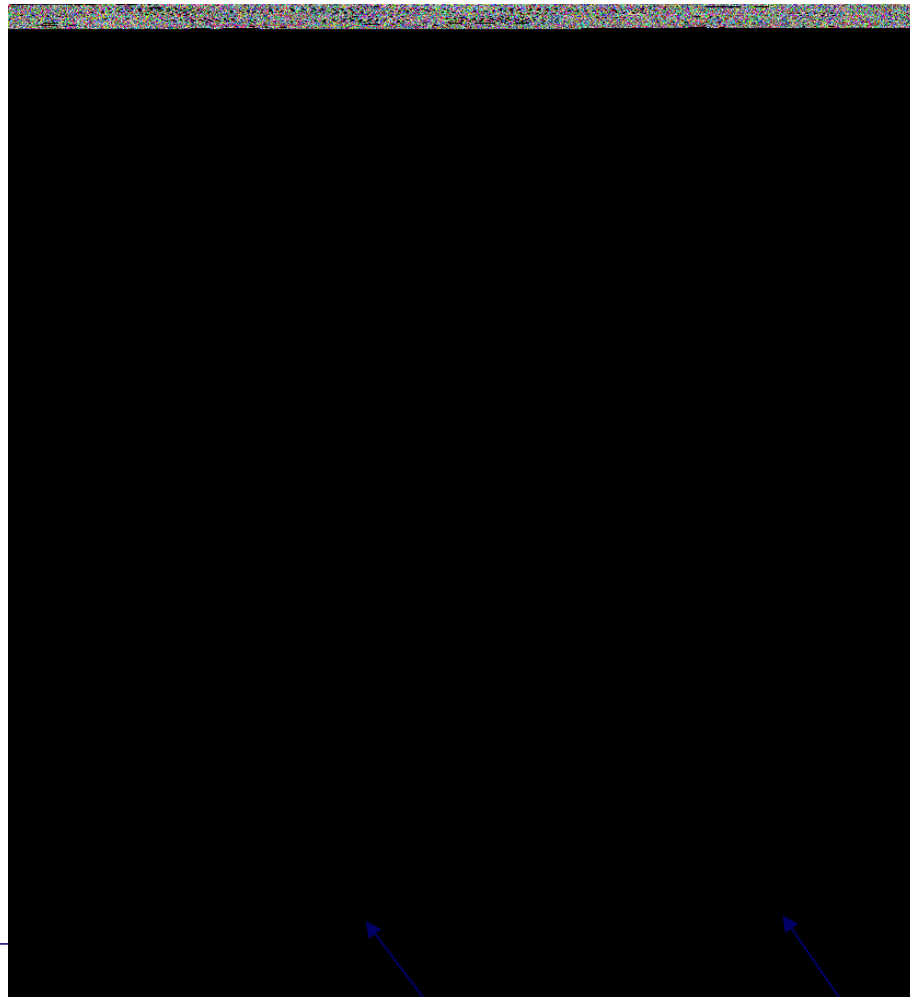


“Auto-Fit” to fit
tension model

GUI Software Tool for Inverse Analysis

11

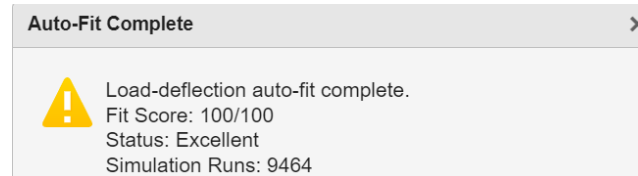
Inverse-Analysis



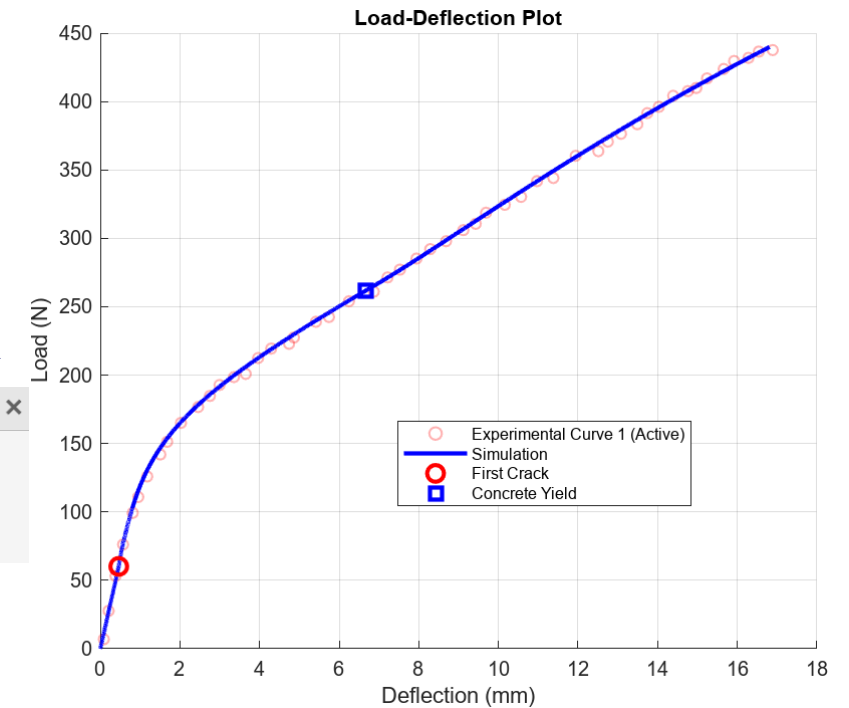
Upload Flexure Data

Click "Auto-Fit"

Auto-fitting



9464 load-deflection
simulations within 10 secs

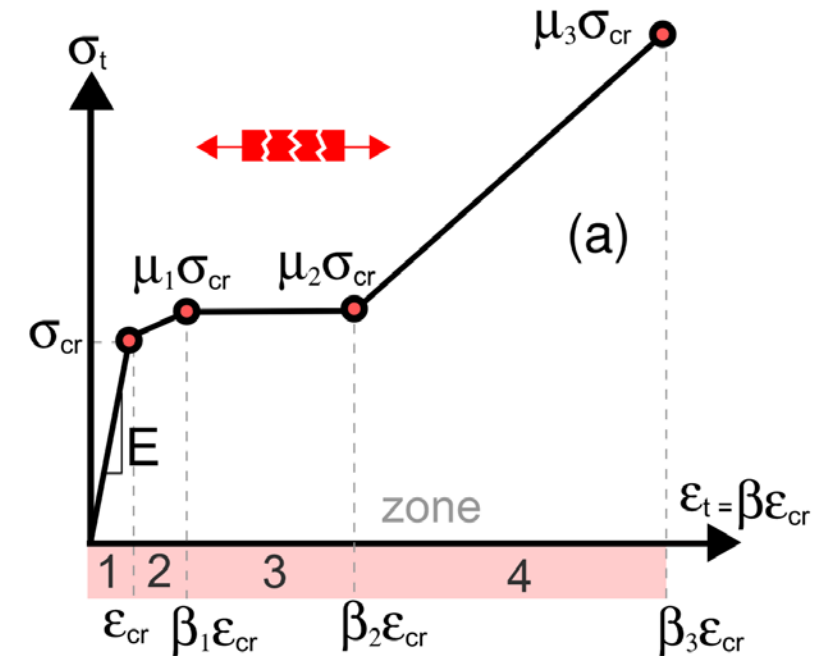


- Closed-Form Analytical Models are highly computational efficient
- Nonlinear least squares with finite difference is used to optimize the tension model parameters

$$S(\beta, \mu) = \sum_{i=1}^n \left[y_i - \hat{y}_i(\beta, \mu) \right]^2$$

Diagram illustrating the components of the equation:

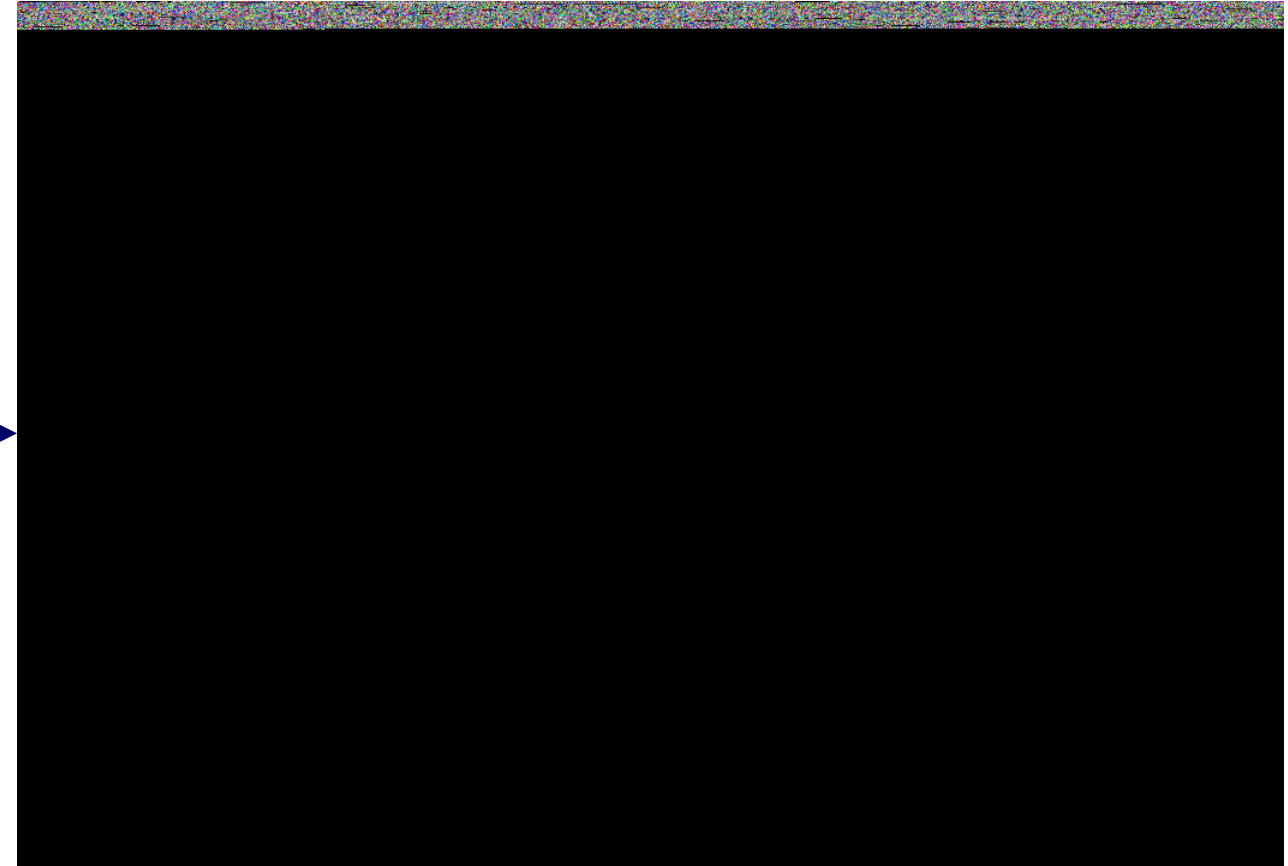
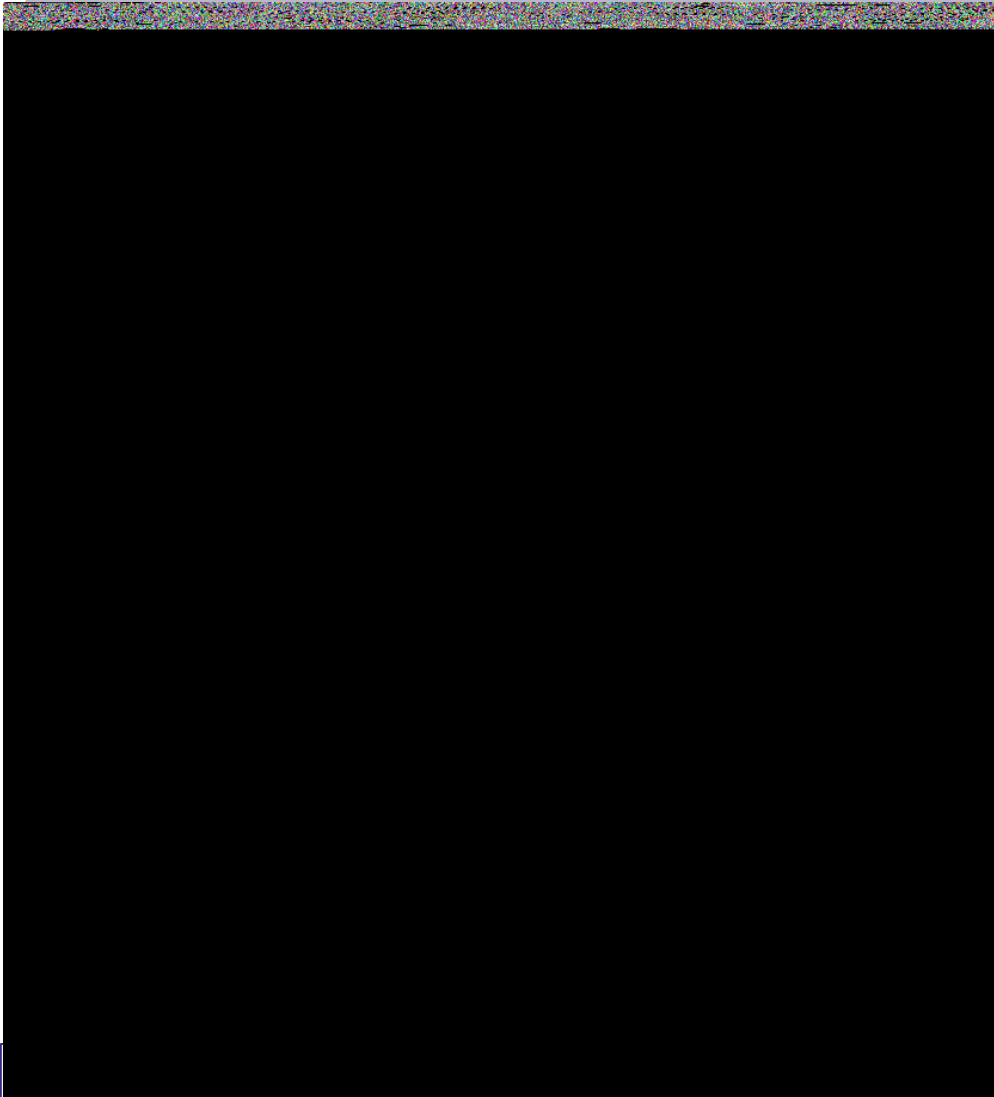
- $S(\beta, \mu)$ is labeled as **Error**.
- y_i is labeled as **Experimental Curve**.
- $\hat{y}_i(\beta, \mu)$ is labeled as **Simulated Curve**.



- n = independent restarts with randomized initial conditions

GUI Software Tool for Inverse Analysis

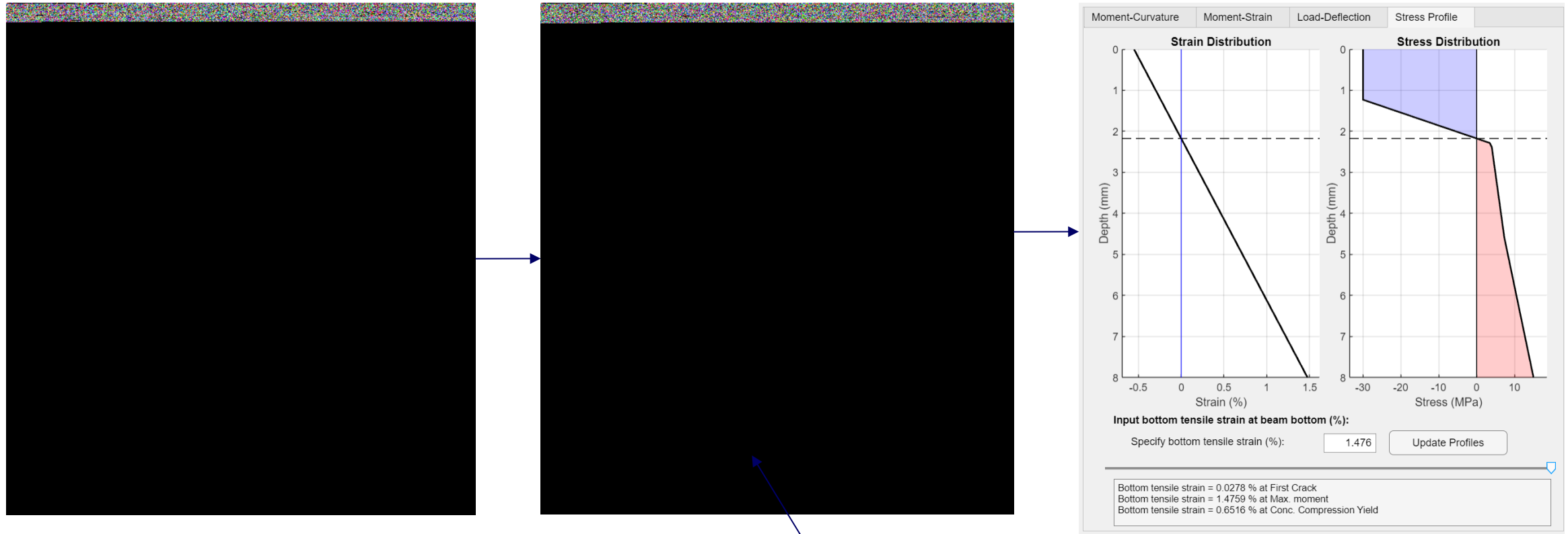
13



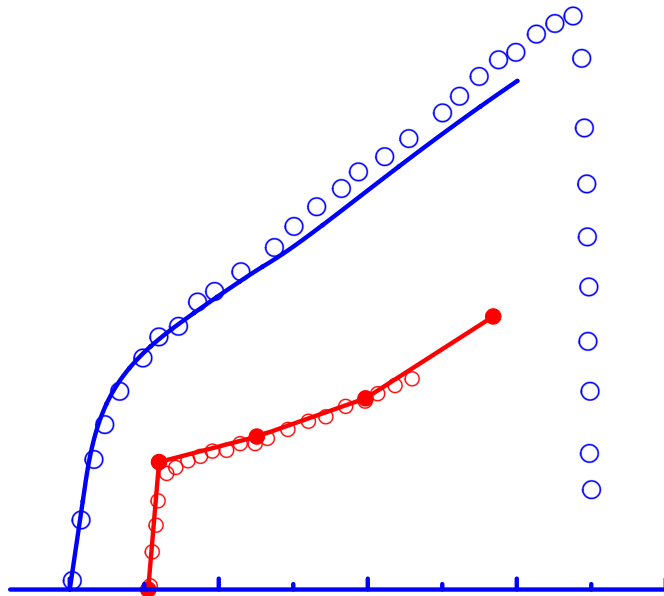
Full Tension Model extraction from Flexure
Inverse-Analysis

GUI Software Tool for Inverse Analysis

14



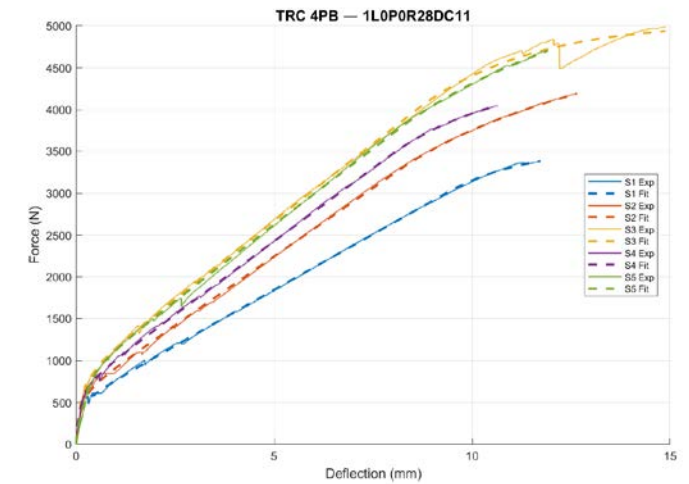
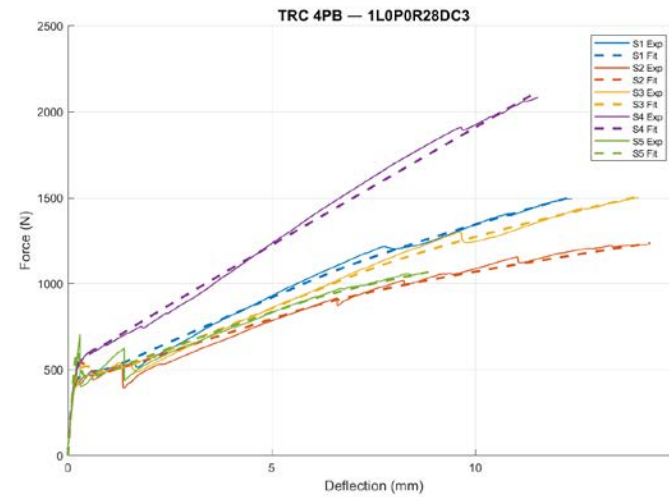
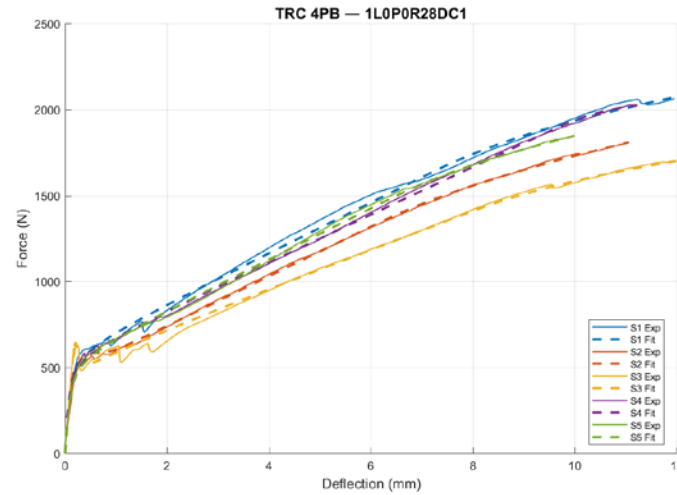
Real-time slider to view stress-distribution



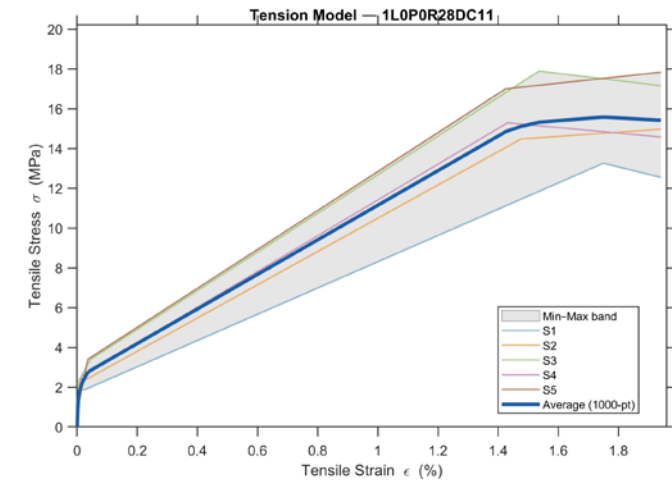
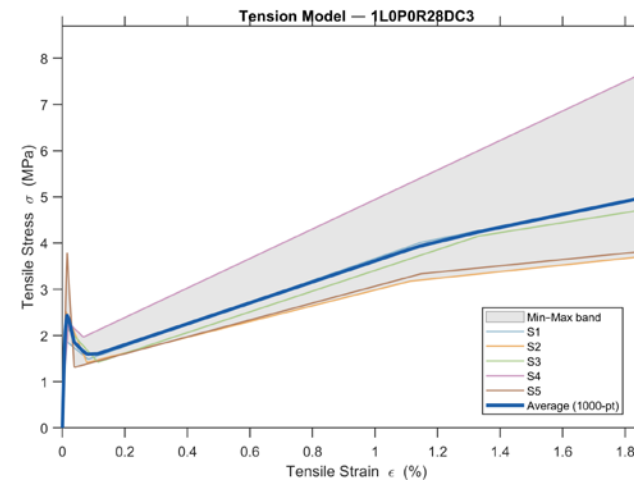
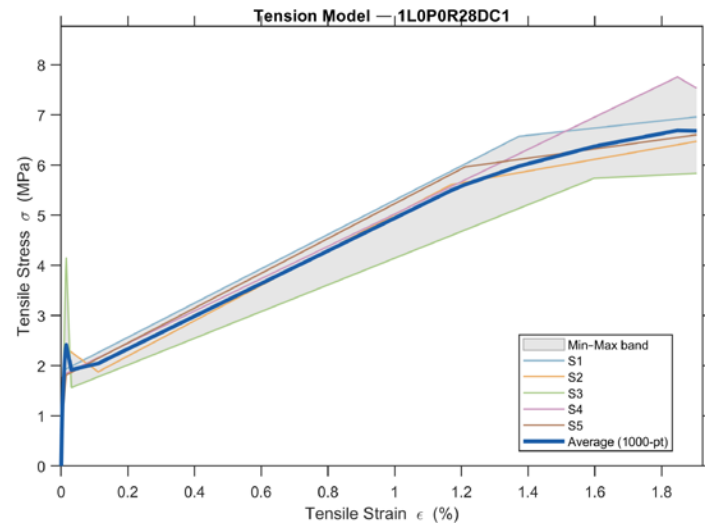
Large Experimental Inverse-Analysis

16

Flexure Data



Back-calculated
Tension Models



M. Hutaibat et al. (2023)

60 Flexure Curves Back-calculated in 5 mins

Code

File available

An Inverse Analysis and Design software for FRC, UHPC, TRC & HRC Cement Composites

February 2025

DOI: [10.13140/RG.2.2.12492.53127/4](https://doi.org/10.13140/RG.2.2.12492.53127/4)

Lab: [Barzin Mobasher's Lab](#)

Devansh Patel

Barzin Mobasher

Chidchanok Pleesudjai

Research Interest Score

Citations

Recommendations

Reads ⓘ

8.0

0

6

479

[Learn about stats on ResearchGate](#)

Overview

Stats

Citations

References

Share

More

Public File ⓘ

Private Files ⓘ

HRC_v4_Pack.zip

6.56 MB · Available from [Devansh Patel](#)

[Download](#) [Remove file](#)

<https://doi.org/10.13140/RG.2.2.12492.53127/4>



Questions?